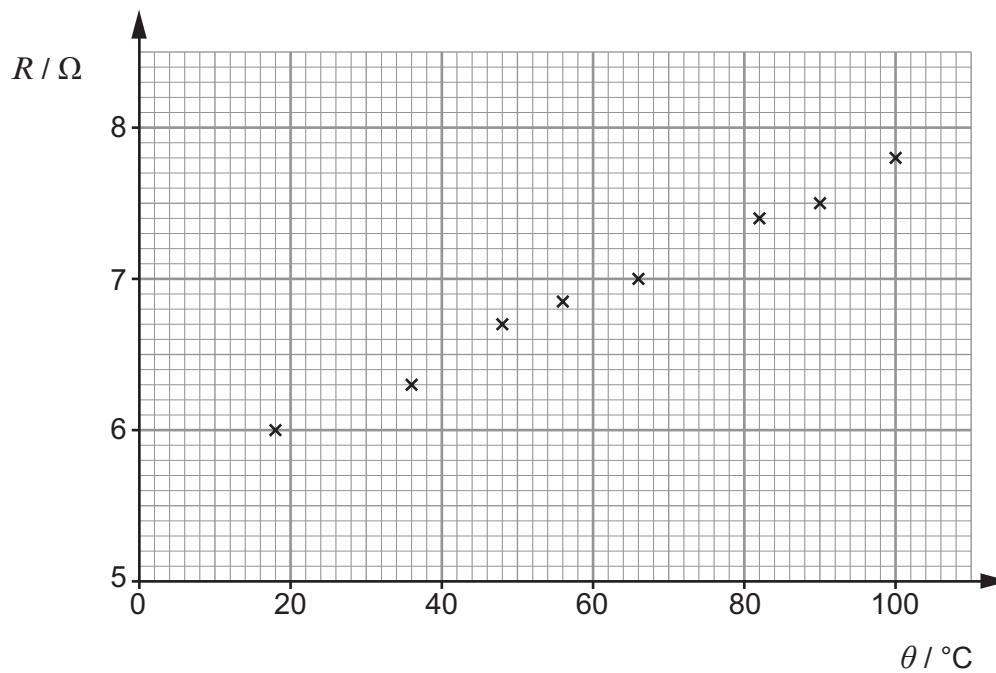


3. A student slowly heated a coil of insulated copper wire. He took readings of its temperature, θ , and resistance, R , at intervals. The readings are plotted below.



- (a) Bearing in mind the range of temperatures, suggest how the student heated the coil, and how he could have extended the range of temperatures down to just above 0°C . [2]

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- (b) Textbooks state that the resistance, R , of a metal wire is related to its celsius temperature, θ , by an equation of the form:

$$R = R_0 \alpha \theta + R_0$$

in which R_0 and α are positive constants.

- (i) Explain how the readings plotted support the equation. [3]

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(ii) Determine from the graph the values, with units, of:

I. R_0 ;

[1]

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II. α .

[3]

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